

June, 2026

## CURRICULUM VITAE AND LIST OF PUBLICATIONS

### Personal Details

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Name - Amit Zeev Dvir, [amitdv@g.ariel.ac.il](mailto:amitdv@g.ariel.ac.il)  
Date and place of birth - 27/07/1976, Israel  
Regular military service (dates) - 1994-97 Paratrooper  
Telephone number at work - +972-74-7233019  
Telephone number at home - +972-52-3447394

### Education

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#### Undergraduate and Graduate Studies

|                  |                                                                                                                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ph.D.            | 2004 – 2010 - Ben-Gurion University of the Negev, Communication Systems Engineering<br>Prof. Michael Segal<br>Optimizing Traffic Parameters in Wireless Network Communication Backbones                        |
| M.Sc.            | 2002 – 2004 - Ben-Gurion University of the Negev - ElectroOptical Engineering<br>Prof. Michael Segal and Dr. Yehuda Ben-Shimol<br>SPLAST: A Novel Approach for Multicasting in Mobile Wireless Ad Hoc Networks |
| B.Sc.<br>Systems | 1998 – 2002, Ben-Gurion University of the Negev - Communication Engineering                                                                                                                                    |

#### Post-Doctoral Studies

|                      |                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2010 -<br>and Rescue | Ariel University, Computer Science, Dr. Boaz Ben Moshe, NETHD                                                                                                |
| 2010-2012            | Budapest University of Technology and Economics,<br>Department of Telecommunications (CrySyS Lab), Dr. Levente Buttyan. Founded by EU HUMAN-MB08-2, Hungary. |

### Academic Ranks and Tenure in Institutes of Higher Education

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|------|-------------------------------------------------------|
| 2023 | Ariel University, Computer and Software Eng., Tenure, |
| 2022 | Ariel University, Computer Science, Associate Prof,   |
| 2019 | Ariel University, Computer Science, Tenure,           |
| 2015 | Ariel University, Computer Science, Senior Lecturer.  |

2012                      The College of Management Academic Studies, Computer Science, Senior Lecturer,

## **Professional Activities**

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### **Positions in academic administration (Departmental, Faculty and University)**

|                |                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 2023 – present | Head of Computer and Software Eng. Department, Ariel University                                                                      |
| 2015 – 2023    | Head of Cyber track, Computer Science, Ariel University                                                                              |
| 2015 – present | Head of Ariel Cyber Innovation Center, Ariel University                                                                              |
| 2015 – 2018    | Head of Student Projects, Computer Science, Ariel University                                                                         |
| 2013 – 2014    | Member of the Committee on Discipline (for Faculty Members),<br>The College of Management Academic Studies                           |
| 2008 – 2014    | Head of Student Projects, Computer Science, The College of<br>Management Academic Studies                                            |
| 2008 – 2014    | Chairman of the Library Committee of the Computer Science<br>School, Computer Science, The College of Management Academic<br>Studies |

### **Editor or member of the editorial board of scientific or professional journal**

- 1) 2026. IEEE Networking Letter
- 2) 2026, IEEE JSTSP, Special Issue on Autonomous and Evolutive Optimization in Networked AI.
- 3) 2022, Guest Editorial, Journal: IET Smart Cities.

### ***Journal Reviewer:***

- 1) 2024 – Present, IEEE Transactions on Cognitive Communications and Networking.
- 2) 2024 – Present, IEEE Communications Magazine.
- 3) 2024 – Present, ACM Computing Surveys.
- 4) 2024 – Present, IEEE Transactions on Networking. 2016 -Present, Telecommunication Systems.
- 5) 2023 – Present, Transactions on Dependable and Secure Computing.
- 6) 2021 – Present, Computers and security
- 7) 2021 – Present, Computer Networks.
- 8) 2020 – present, IEEE Transactions on Emerging Topics in Computational Intelligence.
- 9) 2019 – Present, IEEE/ACM Transactions on Networking.
- 10) 2018 – Present, IEEE Intelligent Transportation Systems Transactions.

11) 2017 – Present, Transactions on Information Forensics & Security.

12) 2016 – Present, Wireless Network.

***Conferences reviewers:***

1) 2016 – Present, IEEE International Conference on Communications

2) 2016 – Present, IEEE Consumer Communications & Networking Conference.

***TPC conferences:***

1) 2027, IEEE Consumer Communications & Networking Conference. Security, Privacy, And Content Protection.

2) 2026, Workshop Co-chair, IEEE WoWMoM, 6nd International Workshop on Smart Computing for Smart Cities

3) 2024, Workshop Co-chair, IEEE WoWMoM, 5nd International Workshop on Smart Computing for Smart Cities

4) 2023, Workshop Co-chair, IEEE WoWMoM, 4nd International Workshop on Smart Computing for Smart Cities

5) 2022, Workshop Co-chair, IEEE WoWMoM, 3nd International Workshop on Smart Computing for Smart Cities

6) 2021, Workshop Co-chair, IEEE WoWMoM, 2nd International Workshop on Smart Computing for Smart Cities

7) 2020, Workshop Co-chair, IEEE WoWMoM, 1nd International Workshop on Smart Computing for Smart Cities

8) 2018, Workshop Co-chair, IEEE WoWMoM, Workshop on Communication and Networking Techniques for Contemporary Video

9) 2017, Workshop Co-chair, IEEE INFOCOM, 3rd Workshop on Communication and Networking Techniques for Contemporary Video

10) 2015, Workshop Co-chair, IEEE INFOCOM, 2nd Workshop on Communication and Networking Techniques for Contemporary Video

**Membership in professional/scientific societies**

2009 – Present            IEEE

2009 – 2015             ACM

## **Educational activities**

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### **Courses taught in Recent Years**

2025 - present, Introduction to Communication Networks, undergraduate, Computer and Software Eng., Ariel University

2025 - present, Introduction to Network Security, undergraduate, Electrical and Electronics Eng., Ariel University

2017 – present, Seminar in advance topics in Cyber Security, undergraduate, Ariel University

2015 – present, Introduction to Communication Networks, undergraduate, Computer Science, Ariel University

2015 – 2019, Introduction to Cyber Security, undergraduate, Ariel University

2013 – 2015, Advance Communication Networks, undergraduate, The College of Management Academic Studies

2013 – 2015, Wireless Communication Networks, undergraduate, The College of Management Academic Studies

### **Supervised Research Students**

#### **Active**

##### **Ph.D.**

- 1) 2025- Present, Zohar Fine, Ph.D, Attacks on LAN, Additional supervisor: Ron Hirschprung
- 2) 2025- Present, Daniel Gilkarov, Ph.D, Attacks on DL, Additional supervisor: Ran Dubin
- 3) 2025- Present, Yedidal Louck, Pre-Ph.D, AI agents attack
- 4) 2024- Present, Natan Delberi, Ph.D, A Novel Approach to Encrypted Traffic Classification using Digital Signal Processing, Additional supervisor: Chen Hajaj
- 5) 2022- Present, Udi Aharon, Ph.D., API security, Additional supervisor: Ran Dubin
- 6) 2022- Present, Jonathan Zion, Ph.D., Classifying GAN, Additional supervisor: Prof Boaz Ben Moshe, Ran Dubin, Chen Hajaj, Ariel University

## **M.Sc**

- 1) 2024- Present, Gilad Livshitz, M.Sc, DoH attacks.
- 2) 2024- Present, Matan Klein, M.Sc., Network classification using GNN,  
Additional supervisor: Ran Dubin
- 3) 2024- Present, Oz Atar, M.Sc., Protocol Reverse Engineering  
using smart agents. Additional supervisor: Ran Dubin

## **Completed**

### **Post-Doc**

- 1) 2022, Nadav Schweitzer, PostDoc, Ariel University, Additional supervisor: Dr.  
Ariel Stulman JCT

### **Ph.D.**

- 1) 2025, Eyal Berliner, Ph.D., Ariel Univeristy Additional supervisor: Prof.  
Shraga Shoval, Ariel University.
- 2) 2022, Aviram Zilberman, Ph.D., Ariel University, Additional supervisor: Dr.  
Yoram Hadad (JCT)
- 3) 2022, Or Anidjar, Ph.D. Ariel University, Additional supervisor: Dr. Chen  
Hajaj, Ariel University
- 4) 2017, Ran Dubin, Ph.D. Ben Gurion University, Additional supervisor: Prof.  
Ofer Hadar BGU

## **M.Sc**

- 1) 2024, Adi Lichy, M.Sc, Ariel University, Additional supervisor: Dr. Chen Hajaj  
Ariel University
- 2) 2023, Porat Aharon, M.Sc, Ariel University, Additional supervisor: Dr. Chen  
HajajAriel University
- 3) 2022, Ofek Beder, M.Sc, Ariel University, Additional supervisor: Dr. Chen  
Hajaj Ariel University

- 4) 2022, Saimon Lankri, M.Sc - Estimating QoE from Encrypted Traffic, Ariel University
- 5) 2022, Liad Choen, M.Sc, Ariel University, Additional supervisor: Dr. Ariel Stulman JCT
- 6) 2020, Nitay Hason - M.Sc, Ariel University, Additional supervisor: Dr. Chen Hajaj Ariel University
- 7) 2019, Nehor Golan – M.Sc, Ariel University, Additional supervisor: Dr. Mira Golan
- 8) 2018, Ronit Ben Shalom – M.Sc., Ariel University
- 9) 2018, Harel Berger – M.Sc. Ariel University
- 10) 2017, Aviram Zilberman – M.Sc, Ariel University, Additional supervisor: Dr. Yoram Hadad (JCT)
- 11) 2017, Roi Inbar – M.Sc, Ariel University, Additional supervisor: Dr. Eran Omri, Ariel University
- 12) 2017, Yehonatan Zion – M.Sc., Ariel University, Additional supervisor: Dr. Ofir Pele
- 13) 2017, Jonathan Muehlstein – M.Sc., Ariel University, Additional supervisor: Dr. Ofir Pele
- 14) 2016, Michel Stewart – M.Sc, Louisiana State University, Additional supervisor: Dr. Rajgopal Kannan (Louisiana State University) .

## **Awards, Citations, Honors, Fellowships**

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### **Honors, Citation Awards (including during studies)**

2015 - College of Management Academic Studies, Encouragement Research award  
2014 - College of Management Academic Studies, Best faculty member award  
2014 - College of Management Academic Studies, Encouragement Research award  
2013 - College of Management Academic Studies, Encouragement Research award  
2013 - College of Management Academic Studies, Best faculty member award  
2013 - Ruppin College - Best lecturer award  
2012 - College of Management Academic Studies, Encouragement Research award  
2010 - College of Management Academic Studies, Best lecturer award  
2003 - Intel - Excellence Award for Higher Academic Degree Studies  
2002 - Teva's Ltd, Excellence Award

### **Fellowships (e.g. Fullbright)**

2010, EU Marie Skłodowska-Curie Fellowships, 100,000\$

## Scientific Publications

### Citation Index

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**H-index (13 / 22):**

**Total number of citations of all articles (668 / 2090):**

**Total number of citations without self-citations (608 / 1183):**

**For all the papers, the researchers have equal contributions, CA (corresponding author), S (student)**

### Conference proceedings

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#### **Publications since joining Ariel University after the Last Promotion**

- 1) Y. Zion (S), E. Paz (S), R. Dubin, **A. Dvir**, C. Hajaj, "Quality of Experience Prediction for First-Person Shooter Online Gaming: The Case Study of Call of Duty", IEEE 23rd Consumer Communications & Networking Conference (CCNC), 2026, **Core=B**. DOI:[10.1109/CCNC65079.2026.11366349](https://doi.org/10.1109/CCNC65079.2026.11366349)
- 2) R. Marbel, Y. Cohen (S), R. Dubin, **A. Dvir**, C. Hajaj, "Cloudy with a Chance of Anomalies: Dynamic Graph Neural Network for Early Detection of Cloud Services' User Anomalies", IEEE 23rd Consumer Communications & Networking Conference (CCNC), 2026, **Core=B**. DOI:[10.1109/CCNC65079.2026.11366510](https://doi.org/10.1109/CCNC65079.2026.11366510)
- 3) Itay Meiri (S), Ran Dubin, **A. Dvir**, Chen Hajaj, "Out-Of-Distribution Is Not Magic: The Clash Between Rejection Rate and Model Success", Conference on Computer Science and Intelligent Systems (IEEE FedCSIS), Kraków, Poland, 14-17 September 2025. **Core=B**. <https://doi.org/10.15439/2025F5708>
- 4) S. Lisker(S), A. Botman, C. Hajaj, R. Dubin, **A. Dvir**. Optimized File Type Detection and One-Shot Retrieval. In Proceedings of the IEEE International Conference on Communications (ICC). 2025. **Core = B**. <https://doi.org/10.1109/ICC52391.2025.11161523>
- 5) Z. Simhon(S), M. Weiss(S), C. Hajaj, R. Marbel, R. Dubin, **A. Dvir**. A New D-MAGIC: Dynamic Model for Cybersecurity Attack Detection Using GNNs into

Clustering. In Proceedings of the IEEE International Conference on Communications (ICC). 2025. **Core = B.** [10.1109/ICC52391.2025.11160734](https://doi.org/10.1109/ICC52391.2025.11160734), extended to J2

- 6) Y. Zion(S), P. Aharon(S), R. Dubin, **A. Dvir**, C. Hajaj. Enhancing Encrypted Internet Traffic Classification Through Advanced Data Augmentation Techniques. In Proceedings of the IEEE International Conference on Communications (ICC). 2025. **Core = B.** [10.1109/ICC52391.2025.11160885](https://doi.org/10.1109/ICC52391.2025.11160885)
- 7) A. Marnerides, C. Hajaj, R. Marbel, R. Dubin, **A. Dvir**. PQClass: Classification of Post-Quantum Encryption Applications in Internet Traffic. In Proceedings of the IEEE International Conference on Communications (ICC). 2025. **Core = B.** [10.1109/ICC52391.2025.11160818](https://doi.org/10.1109/ICC52391.2025.11160818)
- 8) N. Dillbary (S), R. Yozevitch, **A. Dvir**, R. Dubin, C. Haja, "Hidden in Time, Revealed in Frequency: Spectral Features and Multiresolution Analysis for Encrypted Internet Traffic Classification", in Consumer Communications & Networking Conference (CCNC), IEEE Annual. IEEE, 2024, **Core = B.** <https://doi.org/10.1109/CCNC51664.2024.10454807>

#### **Publications since joining Ariel University Before Last Promotion**

- 1) O. Bader(S), A. Lichy(S), C. Hajaj, R. Dubin and **A. Dvir**, "Maldist: From encrypted traffic classification to malware traffic detection and classification," in Consumer Communications & Networking Conference(CCNC), IEEE Annual. IEEE, 2022, **Core = B.** DOI: <https://doi.org/10.1109/CCNC49033.2022.9700625>
- 2) O. H. Anidjar(S), I. Lapidot, C. Hajaj and **A. Dvir**, "A Thousand Words are Worth More Than One Recording: Word-Embedding Based Speaker Change Detection", Interspeech 2021. **Core = A.** <https://doi.org/10.21437/Interspeech.2021-87>
- 3) **A Dvir**, AK Marnerides, R Dubin, N Golan(S), **2019**, "Clustering the Unknown-The Youtube Case", IEEE 2019 International Conference on Computing, Networking and Communications. **Core = B.** <https://doi.org/10.1109/ICCNC.2019.8685599>, extended to J11

- 4) R. Shalala(S), R. Dubin(S), O. Hadar, **A. Dvir**, **2018**, Video QoE Prediction Based on User Profile, International Conference on Computing, Networking and Communications. **Core = B.** <https://doi.org/10.1109/ICCNC.2018.8390245>
  
- 5) **A. Dvir**, Y. Haddad, A. Zilberman(S), **2018**, Wireless SDN Controller Placement Problem, IEEE Consumer Communications and Networking Conference, **Core = B.** <https://doi.org/10.1109/CCNC.2018.8319228>, extend J14
  
- 6) R Dubin, **A Dvir**, O Pele , J Muehlstein (S),Y. zion(S), M. Bahumi(S), I Kirshenboim(S), **2017**, Analyzing HTTPS Encrypted Traffic to Identify User's Operating System, Browser and Application, IEEE Consumer Communications and Networking Conference, **Core = B.** <https://doi.org/10.1109/CCNC.2017.7983154>
  
- 7) M. Stewart(S), **A. Dvir**, R. Kannan, B. Krishnamachari, **2016**, CASPaR: Congestion Avoidance Shortest Path Routing for Delay Tolerant Networks, International Conference on Computing, Networking and Communications, **Core = B.** <https://doi.org/10.1177/1550147717741264>
  
- 8) R. Dubin(S) , **A. Dvir** , O. Hadar, R. Shalala(S), O. Ahark, **2015**, Video Complexity Hybrid Traffic Shaping for HTTP Adaptive Streaming, IEEE Consumer Communications and Networking Conference, **Core = B.** <https://doi.org/10.1109/CCNC.2015.7158061>
  
- 9) R. Dubin(S) , **A. Dvir** , O. Hadar, T. Frid(S) and A. Vesker (S) **2015**, Novel Ad Insertion Technique for MPEG-DASH, IEEE Consumer Communications and Networking Conference, **Core = B.** <https://doi.org/10.1109/CCNC.2015.7158038>

### **Publications before arrived to Ariel University**

- 1) B. Ben-Moshe, A. Dvir, E. Berliner(S), **2013**, Performance Monitoring Framework for Wi-Fi MANET IEEE Wireless Communications and Networking Conference pp. 4463-4468. **Core = B.** <https://doi.org/10.1109/WCNC.2013.6555297>
- 2) Dvir, L. Buttyan, V. T. Ta, **2013**(S), SDTP+: Securing a Distributed Transport Protocol for WSNs using Merkle Trees and Hash Chains IEEE International Conference on Communications pp. 2073-2078. **Core = B.** <https://doi.org/10.1109/ICC.2013.6654831>, extended J19
- 3) R. Dubin(S), O. Hadar, A. Dvir, **2013**, The Effect of Client Buffer and MBR Consideration on DASH Adaptation Logic, IEEE Wireless Communications and Networking Conference, pp. 2178-2183. **Core = B.** <https://doi.org/10.1109/WCNC.2013.6554900>
- 4) B. Ben-Moshe, A. Branderud(S) and A. Dvir, **2012**, Modeling adaptive rate video transmission in Wi-Fi MANET ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems, pp. 167-174. **Core = A** [.https://doi.org/10.1145/2387191.2387214](https://doi.org/10.1145/2387191.2387214)
- 5) A. Dvir and L. Buttyan, **2011**, Backpressure Approach for Bypassing Jamming Attacks in Wireless Sensor Networks IEEE INFOCOM. (**167 citations**) **Core= A\***. [DOI-research](https://doi.org/10.1109/INFOCOM.2011.6235444)
- 6) A. Dvir, T. Holczer(S) and L. Buttyan, **2011**, VeRA - Number and Rank Authentication in RPL Workshop on Wireless and Sensor Networks Security, IEEE International Conference on Mobile Ad-hoc and Sensor Systems, pp. 709-714 (**299 citations**). **Core = B.** <https://doi.org/10.1109/MASS.2011.76>
- 7) B. Ben-Moshe and A. Dvir, **2011**, Development and optimization of cache element for access layer Work in Progress (Short Papers), IEEE Consumer Communications and Networking Conference pp 1126 – 1127. **Core = B.** <https://doi.org/10.1109/CCNC.2011.5766350>

- 8) Ben-Moshe, **A. Dvir**, M. Segal, and A. Tamir, **2010**, Centdian Computation for Sensor Networks Theory and Applications of Models of Computation, pp. 187-198. **Core = B.** [https://doi.org/10.1007/978-3-642-13562-0\\_18](https://doi.org/10.1007/978-3-642-13562-0_18), extend J23
  
- 9) **A. Dvir**, A. V. Vasilakos, **2010**, Backpressure-based Routing Protocol for DTN, ACM SIGCOMM (167 citations). **Core = A\***. <https://doi.org/10.1145/1851182.1851233>
  
- 10) **A. Dvir**, and N. Carlsson, **2009**, Power-aware Recovery for Geographic Routing IEEE Wireless Communications & Networking Conference pp. 1-6. **Core = B.** <https://doi.org/10.1109/WCNC.2009.4917915>
  
- 11) **A. Dvir**, and M. Segal, **2007**, Placing and Maintaining a Core Node in Wireless Ad Hoc Sensor Networks IFIP Networking, 13-24. **Core= A.** [https://doi.org/10.1007/978-3-540-72606-7\\_2](https://doi.org/10.1007/978-3-540-72606-7_2), extended J24
  
- 12) B. Ben-Moshe(S), Y. Ben-Yehezkel(S), Y. Ben-Shimol, **A. Dvir**, and M. Segal, **2006** An Automated Wireless Fixed-Access Network Antenna Positioning Algorithm, IEEE Consumer Communications and Networking Conference, 902 – 906. **Core = B.** <https://doi.org/10.1109/CCNC.2006.1593169>
  
- 13) Y. Ben-Shimol, **A. Dvir**, and M. Segal, **2004**, SPLAST: A Novel Approach for Multicasting in Mobile Wireless Ad Hoc Networks, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), 1011 – 1015. **Core = B.** <https://doi.org/10.1109/PIMRC.2004.1373851>

## Articles

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**For all the papers, the researchers have equal contributions, CA (corresponding author, in our case it is the student most of the cases), S (student)**

### **Publications since joining Ariel University after Last Promotion**

- 1) Y Louck(S, CA), A Dvir, A Stulman, “[Security Analysis of Agentic AI Communication Protocols: A Comparative Evaluation](#)”, ACM Transactions on AI Security and Privacy, 2026. <https://doi.org/10.1145/3803431>
- 2) Z. Simhon (S, CA), M. Weiss(S), R. Marbel, C. Hajaj, A. Dvir, R. Dubin, “Real-Time Network Security: Integrating ANN and Dynamic Graph-Based Clustering”, Computer Networks, Volume 227, 2026. **IF=4.6, Citation = 0**, JR=58/258, Q1 <https://doi.org/10.1016/j.comnet.2026.112016>, conference version c5
- 3) A Zilberman (S, CA), A Dvir, A Stulman, “IPv6 Routing Protocol for Low-Power and Lossy Networks Security Vulnerabilities and Mitigation Techniques: A Survey”. ACM Computing Surveys 57 (11), 1-77, **IF=28, Citation = 0**, JR=1/147, Q1, 2025. <https://doi.org/10.1145/3732776>
- 4) N. Schweitzer (S, CA), L. Cohen (S), T. Hirst, A. Dvir, A. Stulman, "Achieving manet protection without the use of superfluous fictitious nodes", Computer Communications 229, **IF=4.5, Citation = 3**, JR=51/249, Q1, 2025. <https://doi.org/10.1016/j.comcom.2024.107978>
- 5) U. Aharon (S, CA), R. Dubin, A. Dvir, C. Hajaj. A Classification-by-Retrieval Framework for Few-Shot Anomaly Detection to Detect API Injection Attacks. In Computers & Security. JR=46/249, **IF = 4.8, citations = 5**, Q1, 2025. <https://doi.org/10.1016/j.cose.2024.104249>
- 6) M. Sidorov (S, CA), R. Birman (S), O. Hadar, A. Dvir. "Estimating QoE from Encrypted Video Conferencing Traffic", Sensors 25 (4), 1009, **IF=3.4, Citation = 1**, JR=122/353, Q2, 2025. <https://doi.org/10.3390/s25041009>

- 7) O. H. Anidjar (S, CA), R. Marbel, R. Dubin, **A. Dvir**, C. Hajaj, “Extending limited datasets with GAN-like self-supervision for SMS spam detection”, Computers & Security, **IF=4.8, Citation = 4**, JR=46/249, Q1, 2024.  
<https://doi.org/10.1016/j.cose.2024.103998>
  
- 8) C. Hajaj, P. Aharon (S, CA), R. Dubin, **A. Dvir**, “The art of time-bending: Data augmentation and early prediction for efficient traffic classification”, Expert Systems with Applications, 252, **IF=7.5, Citation = 2**, JR=24/197, Q1, 2024.  
<https://doi.org/10.1016/j.eswa.2024.124166>
  
- 9) A. Zilberman(S, CA), **A. Dvir**, A. Stulman, “SPRINKLER: A Multi-RPL Man-in-the-Middle Identification Scheme in IoT Networks”, IEEE Transactions on Mobile Computing, **IF=7.7, Citation = 6**, JR=11/249, Q1, 2024.  
<https://doi.org/10.1109/TMC.2024.3370894>
  
- 10) O. Bader(S, CA), A. Lichy (S), **A. Dvir**, R. Dubin, C. Hajaj, “OSF-EIMTC: An open-source framework for standardized encrypted internet traffic classification”, Computer Communications, 213, 271-284, **IF=4.5, Citation = 1**, JR=51/249, Q1, 2024.  
<https://doi.org/10.1016/j.comcom.2023.10.011>
  
- 11) N. Schweitzer(S, CA), **A. Dvir**, A. Stulman, “Network wormhole attacks without a traditional wormhole”, Ad Hoc Networks, 151, 103286, **IF=4.4, Citation = 4**, JR=53/249, Q1, 2023. <https://doi.org/10.1016/j.adhoc.2023.103286>
  
- 12) H. Berger(S, CA), **A. Dvir**, E. Mariconti, C. Hajaj, “Breaking the structure of MaMaDroid”, Expert Systems with Applications, 228, 120429, **IF=8.039, Citation = 2**, JR=21/144, Q1, 2023. <https://doi.org/10.1016/j.eswa.2023.120429>
  
- 13) N. Schweitzer(S, CA), L. Cohen (S), **A. Dvir**, A. Stulman, “Persuasive: A node isolation attack variant for OLSR-based MANETs and its mitigation”, Ad Hoc Networks, 148, 103192, 1-8, **IF=4.4, Citation = 5**, JR=53/249, Q1, 2023.  
<https://doi.org/10.1016/j.adhoc.2023.103192>
  
- 14) U. Ben-Shalom, R. Mash, **A. Dvir**, E. Lewin, “Female violence in terror attacks: a phenomenological analysis based on evidence from the Intifada of the Individuals”,

Crime, Law and Social Change, 80 (2), 173-194, **IF=1.4, Citation** = 1, JR=124/263, Q2, 2023. <https://doi.org/10.1007/s10611-023-10076-z>

15) A. Zilberman(S, CA), A. Stulman, **A. Dvir**, “Identifying a malicious node in a uav network”, IEEE Transactions on Network and Service Management, Volume: 21 (1), 1226 – 1240, **IF=4.7, Citation** = 10, JR=48/249, Q1, 2023. <https://doi.org/10.1109/TNSM.2023.3300809>

16) A. Treistman(S, CA), D. Mughaz, A. Stulman, **A. Dvir**, “Word embedding dimensionality reduction using dynamic variance thresholding (DyVaT)”, Expert Systems with Applications, 208, pp 118-157, **IF=8.039, Citation** = 13, JR=21/144, Q1, 2022. <https://doi.org/10.1016/j.eswa.2022.118157>

17) O. H. Anidjar(S, CA), Y. Est`eve, C. Hajaj, A. Dvir, I. Lapidot, “Speech and Multilingual Natural Language Framework for Speaker Change Detection and Diarization”, Expert Systems with Applications, 213, 119-238, **IF=8.039, Citation** = 10, JR=21/144, Q1, 2022. <https://doi.org/10.1016/j.eswa.2022.119238>

#### **Publications since joining Ariel University and before Last Promotion**

1) A. Lichy(S, CA), O. Bader (S), R. Dubin, A. Dvir, C. Hajaj, “When a RF Beats a CNN and GRU, Together-A Comparison of Deep Learning and Classical Machine Learning Approaches for Encrypted Malware Traffic Classification”, Computers & Security, 124, 103000, **IF=5.29, Citation** = 50, JR=42/164, Q2, 2022. <https://doi.org/10.1016/j.cose.2022.103000>

2) C. Hajaj, N. Hason(S, CA), **A. Dvir**, “Less is more: Robust and novel features for malicious domain detection”, Electronics 11 (6), 969. **IF=2.408, Citation** = 29, JR=93/161, Q3, 2022. <https://doi.org/10.3390/electronics11060969>

3) H. Berger(S, CA), C. Hajaj, E. Mariconti and, **A. Dvir**, “Crystal Ball: From Innovative Attacks to Attack Effectiveness Classifier”, IEEE Access, **IF=4.076, Citation** = 9, JR=35/156, Q1, 2021. <https://doi.org/10.1109/ACCESS.2021.3138628>

4) A. Zilberman(S, CA), Y. Haddad, S. Erlich (S), Y. (Joseph) Peretz and **A. Dvir**, 2021, “Heterogeneous SDN Controller Placement Problem - The Wi-Fi and 4G LTE-U

- Case”, Computer Networks, **IF=3.78, Citation = 5**, JR=4/32, Q1  
<https://doi.org/10.1016/j.comnet.2021.108376>
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  - 8) U. Ben Shalom, **A. Dvir**, R. Mash, R. Moshe, 2020, Micro-Sociology and New Wars - Visual analysis of terror attacks during the Intifada of the Individuals, Armed Forces and Society, **IF = 0.918, Citation = 15**, JR=102/150 , Q3.  
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  - 9) Harel Berger(S, CA), **A. Dvir**, Moti Geva, 2020, A wrinkle in time: a case study in DNS poisoning, International Journal of Information Security. **IF = 1.871, Citation = 16**, JR= 53/108 , Q2. <https://doi.org/10.1007/s10207-020-00502-x>
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 5, JR=40/150 , Q2. <https://doi.org/10.1177/1012690217715298>
  
- 13) Uzi Ben-Shalom, **Amit Dvir**, Nadav Gabay(S), Moti Zwilling, Moshe Levy, Eithan Orkibi, 2019, Can fans’ social media activity before the game predict players’ aggression during the game? Evidence from Israeli Basketball Premier League, Sport in Society, 1-9. **IF = 0.939, Citation = 5**, JR= 101/150, Q3.  
<https://doi.org/10.1080/17430437.2019.1637423>
  
- 14) **A. Dvir**, Y Haddad, A Zilberman(S, CA), 2019, The controller placement problem for wireless SDN, Wireless Networks, **IF = 2.357, Citation = 36**, JR=72/156 , Q2.  
<https://doi.org/10.1007/s11276-019-02077-5>, conference version c5 (before promotion)
  
- 15) **A. Dvir**, N Harel, R Dubin(S, CA), R Barkan, R Shalala (S), O Hadar, 2019, MiSAL-A minimal quality representation switch logic for adaptive streaming, Multimedia Tools and Applications, **IF = 2.313, Citation = 5**, JR=34/108 , Q2.  
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- 16) R. Dubin, R. Shalala (S), **A. Dvir**, O. Pele, O. Hadar, 2019, A Fair Server Adaptation Algorithm for HTTP Adaptive Streaming using Video Complexity, Multimedia Tools and Applications, **IF = 2.313, Citation = 10**, JR=34/108 , Q2.  
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- 17) R Dubin, **A Dvir**, O Pele, O Hadar, I Katz(S), O Mashiach, 2018, Adaptation Logic for HTTP Dynamic Adaptive Streaming using Geo-Predictive Crowdsourcing", Multimedia Systems, **IF = 1.734, Citation = 88**, JR=49/108 , Q2.  
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- 18) R. Dubin, O. Hadar, **A. Dvir**, O. Pele, 2018, Video Quality Representation Classification of Encrypted HTTP Adaptive Video Streaming, KSII Transactions on Internet and Information Systems, **IF = 0.626, Citation = 8**, JR=151/156 , Q4.  
<http://doi.org/10.3837/tiis.2018.08.014>

- 19) A. **Dvir**, VT. Ta, S. Erlich(S), L. Buttyan, 2018, STWSN: A Novel Secure Distributed Transport Protocol for Wireless Sensor Networks, International Journal of Communication Systems, **IF = 1.137**, **Citation = 14**, JR=198/266 , Q3. <https://doi.org/10.1002/dac.3827>, conference version c2 (before Ariel)
- 20) R. Dubin(S, CA), A. **Dvir**, O. Pele, and O. Hadar, **2017**, I Know What You Saw Last Minute-Encrypted HTTP Adaptive Video Streaming Title Classification, IEEE Transactions on Information Forensics and Security, 12 (12), 3039-3049 **IF = 6.289**, **Citation = 126**, JR=9/108 , Q1. <https://doi.org/10.1109/TIFS.2017.2730819>
- 21) M. Stewart(S, CA), A. **Dvir**, R. Kannan, B. Krishnamachari, 2017, CASPaR: Congestion Avoidance Shortest Path Routing for Delay Tolerant Networks, International Journal of Distributed Sensor Networks, 13 (11), **IF = 1.236**, **Citation = 24**, JR=129/156 , Q4. <https://doi.org/10.1177/1550147717741264>

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- 1) R. Yozevitch(S, CA), B. Ben-Moshe, A. **Dvir**, 2014, GNSS Accuracy Improvement Using Rapid Shadow Transitions, IEEE Transactions on Intelligent Transportation Systems, Vol. 15, Num. 3, pp. 1113-1122. **IF = 6.709**, **Cite = 49**, JR= 3/36, Q1. <https://doi.org/10.1109/TITS.2013.2294537>
- 2) Ben-Moshe, A. **Dvir**, M. Segal, and A. Tamir, 2012, Centdian Computation in Cactus Graphs, Journal of Graph Algorithms and Applications, Vol. 16, Num. 2, pp. 199 – 224, **IF = 1.00**, **Cite = 32**, Q2. conference version c5 (before Ariel)
- 3) A. **Dvir**, and M. Segal, 2010, Placing and Maintaining a Core Node in Wireless Ad Hoc Sensor Network, Journal of Wireless Communications and Mobile Computing, 10(1), 826-842, **IF = 1.445**, **Cite = 8**, JR=101/156, Q3. [https://doi.org/10.1007/978-3-540-72606-7\\_2](https://doi.org/10.1007/978-3-540-72606-7_2), conference version c11 (before Ariel)
- 4) B. Ben-Moshe, Y. Ben-Yehzekel, Y. Ben-Shimol, A. **Dvir**, and M. Segal 2007, An Automated Wireless Fixed-Access Network Antenna Positioning, Algorithm Journal of Heuristics 13(3): 243-263, **IF = 2.04**, **Cite = 26**, JR=47/108, Q2. <https://doi.org/10.1007/s10732-007-9015-5>

## Refereed articles and refereed letters in scientific journals in HEBREW, running numbers

### Publications since arrived to Ariel University

1. בן שלום, ע., משה, ר., משה, ר., and דביר, ע. 2021. מיקרו-סוציולוגיה של מלחמות חדשות / פרופ' עוזי בן-שלום, רינת משה, ד"ר רוני משה, ד"ר עמית דביר. חברה, צבא וביטחון לאומי 2 (חשוון תשפ"ב, אוקטובר 2021), עמ' 67-87.
2. Ben-Shalom, U., Moshe, R., Mash, R., & **Dvir, Z. Amit.** (2021). Micro-Sociology and New Wars - Visual analysis of terror attacks during the "Intifada of the Individuals. *Hevra, Zava & Bithon Leumy*, 1(2): 67-87(**in Hebrew – Located in Haifa Index**).
3. בן שלום, ע., גבאי, נ., לוי, מ., דביר, ע., אורקיבי, א., and צוילינג, מ. 2021. רגשות חיוביים ושליילים בשיח אוהדים ברשתות חברתיות כמנבאים אלימות של שחקנים במגרשי ספורט - ממצאים מתחומי הכדורגל והכדורסל / ד"ר עוזי בן-שלום, ד"ר נדב גבאי, ד"ר משה לוי, ד"ר עמית דביר, ד"ר איתן אורקיבי וד"ר מוטי צוילינג. מגמות: רבעון למדעי ההתנהגות 56, 1 (אייר תשפ"א, אפריל 2021), עמ' 26-43.
4. Ben-Shalom, U., Gabay, N., Levi, M., **Dvir, A.**, Orkibi, E., & Zwillling, M. (2021). Fans' positive and negative sentiments on social media as predictors of players' aggression in Football and Basketball games: Evidence from Israeli Premier Leagues. *Megamot*, 46(1), 26-43. (**in Hebrew – Located in Haifa Index**).
5. לוי, מ. (סוציולוג), **דביר, ע.**, בן שלום, ע., אורקיבי, א., צוילינג, מ., and גבאי, נ. 2019. שיח אוהדים מקוון כמנבא של אלימות במגרשי כדורגל / ד"ר משה לוי, ד"ר עמית דביר, ד"ר עוזי בן שלום, ד"ר איתן אורקיבי, ד"ר מוטי צוילינג, ד"ר נדב גבאי. רוח הספורט 5 (תשע"ט, 2019), עמ' 1-14.
6. Levy, M. **Dvir, A.** Ben Shalom, U. Zwillling, M., Orkibi, E., Gabay, N. & Pele, O. (2019). Soccer is played on the weekends while violence goes on the whole week – Discourse of violence in Israeli soccer. *Roh Hasport*, 5, 1-14. (**in Hebrew – Located in Haifa Index**).

## Other publication

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- 1) N. Hason(S), **A. Dvir**, and C. Hajaj. "Robust Malicious URL Detection". In the 4th International Symposium on Cyber Security Cryptology and Machine Learning (CSCML 2020). **Core = None.** [https://doi.org/10.1007/978-3-030-49785-9\\_4](https://doi.org/10.1007/978-3-030-49785-9_4)
- 2) N. Berger(S), **A. Dvir**, and C. Hajaj. "Evasion Is Not Enough: A Case Study of Android Malware. In the 4th International Symposium on Cyber Security Cryptology and Machine Learning (CSCML 2020). **Core = None.** [https://doi.org/10.1007/978-3-030-49785-9\\_11](https://doi.org/10.1007/978-3-030-49785-9_11)
- 3) VT. Ta, **A. Dvir**, Y. Arie(S), **2018**, Securing Road Traffic Congestion Detection by Incorporating V2I Communications IEEE WoWMoM workshop, **Core = None.** <https://doi.org/10.1109/WoWMoM.2018.8449746>
- 4) R Dubin(S), **A Dvir**, O Pele, O Hadar, **2016**, I Know What You Saw Last Minute- Encrypted HTTP Adaptive Video Streaming Title Classification, Blackhat London. **Core = None. No DOI**
- 5) R Dubin(S), **A Dvir**, O Pele, O Hadar, **2016**, Real Time Video Quality Representation Classification of Encrypted HTTP Adaptive Video Streaming-the Case of Safari, IEEE The Digital Media Industry and Academic Forum, **Core = None, No DOI**
- 6) R. Dubin(S), **A. Dvir**, O. Hadar, Y. Freifeld(S), A. Ruham, N. Harel, R. Barkan, **2015**, Hybrid Clustered Peer-Assisted DASH-SVC System, Third International Workshop on Multimedia Communications and Networking, **Core = None.** <https://doi.org/10.1109/CIT/IUCC/DASC/PICOM.2015.249>
- 7) R. Dubin(S), **A. Dvir**, O. Hadar, N. Harel, R. Barkan, **2015**, Multicast adaptive logic for Dynamic Adaptive Streaming over HTTP network, INFOCOM-Workshop, 269-274, **Core = None.** <https://doi.org/10.1109/INFCOMW.2015.7179396>

**Publications before arrived to Ariel University**

- 1) **Dvir**, L. Buttyan, V. T. Ta(S), **2014**, Formal Security Verification of Transport Protocols for Wireless Sensor Networks, International Conference on Computer Science, Applied Mathematics and Applications (ICCSAMA), pp. 389-403. **Core=** **None**. [https://doi.org/10.1007/978-3-319-06569-4\\_29](https://doi.org/10.1007/978-3-319-06569-4_29)
- 2) R. Yozevitch(S), B. Ben-Moshe, **A. Dvir**, **2013**, Accurate Lane Detection Using Commercial GNSS Devices Workshop on Positioning, Navigation and Communications, pp. 1-5. **Core=****None**. <https://doi.org/10.1109/WPNC.2013.6533289>
- 3) B. Ben-Moshe, **A. Dvir** and A. Solomon(S), **2011**, Analysis and Optimization of Live Streaming for Over The Top Video, Workshop on Emerging Densely Connected Network, IEEE Consumer Communications and Networking Conference pp. 60-64. **Core =****None**. <https://doi.org/10.1109/CCNC.2011.5766553>
- 4) B. Ben-Moshe, E. Berliner(S), **A. Dvir** and A. Gorodischer(S), **2011**, A joint framework of passive monitoring system for complex wireless networks Workshop on Emerging Densely Connected Network, IEEE Consumer Communications and Networking Conference, pp. 55-59. **Core =****None**. <https://doi.org/10.1109/CCNC.2011.5766543>
- 5) **A. Dvir**, and M. Segal, **2008**, The (k, l) Coredian Tree for Ad Hoc Networks Conference on Distributed Computing Systems Workshops, 267-272. **Core =** **None**. <https://doi.org/10.1109/ICDCS.Workshops.2008.32>
- 6) T. V. Thong(S, CA), L. Buttyán, **A. Dvir**, 2014, On formal and automatic security verification of WSN transport protocols, ISRN Sensor Networks, **IF = no impact, Cite = 3**, no DOI
- 7) **A. Dvir**, R. Giladi, I. Kitroser and M. Segal, 2010, Efficient Decision Handoff Mechanism for Heterogeneous Networks, International Journal of Wireless & Mobile Networks, Vol. 2, Num. 1, pp. 1-13, **IF = no impact, Cite = 20**, no DOI
- 8) Y. Ben-Shimol, **A. Dvir**, and M. Segal, 2008. SPLAST: A Novel Approach for Multicasting in Mobile Wireless Ad Hoc Networks", Advances in Wireless Networks

and Mobile Computing, pp. 256-273. Springer, **Cite = 27, IF= no impact.**

[https://doi.org/10.1007/978-0-387-68567-0\\_11](https://doi.org/10.1007/978-0-387-68567-0_11)

- 9) A. **Dvir**, and M. Segal, 2008, The (k, l) Coredian Tree for Ad Hoc Networks, Ad Hoc & Sensor Wireless Networks, 6(1-2): 123-144, **IF = 0.7, Cite = 6, JR=142/156 , Q4. no DOI**
- 10) B. Ben-Moshe, **A. Dvir**, H. Levi(S), P. Shamil(S), **2013**, Indoor Positioning using Time of Flight Fingerprinting of Ultrasonic Signals International Conference on Indoor Positioning and Indoor Navigation (IPIN), pp. 1-6. **Core = None. No DOI**

## Patents and CVEs

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### Ariel University Patents: 3

1. 2024, Eyal Paz, Ran Dubin, **Amit Dvir**, Chen Hajaj, First-Person Shooter Online Gaming QoE Prediction from Network Traffic Analysis, [US Application No. 63/666,781]
2. 2023, Ran Dubin, **Amit Dvir**, Udi Aharon, System and Method of Application Programming Interface (API) Anomaly Detection [Israeli Patent Application No. 300083]
3. 2022, Ran Dubin and **Amit Dvir**, System and Method for Zero-Trust Application Programming Interface Security [ US Provisional Patent Submission No. 63/346097]
4. CVE-2026-30968
5. CVE-2026-30969
6. CVE-2026-30970

## Research Grants

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### Competitive Research

- 1) 2024 – 2026, MOST, PQC classification, Amit Dvir and Chen Hajaj, 150,00\$
- 2) 2023 – 2024, The Ministry of Regional Cooperation, 20,000\$.

### Commercial Research

- 1) 2025 – 2026, MAFAT, identify network devices, Amit Dvir and Ran Dubin 200,000\$
- 2) 2024 – 2025, Israel Innovation Authority, Optimizing Network classification, Amit Dvir and Ran Dubin 200,000\$
- 3) 2021 – 2024, Israel Innovation Authority, Magnet, ENTM, Amit Dvir, SDN-WCP, 300,000\$
- 4) 2021 – 2023, Israel Innovation Authority, Magnet, Encrypted network traffic - Amit Dvir, SDN-WCP, 200,000\$
- 5) 2019 – 2021, Israel Innovation Authority, Magnethon, Flash Networks ltd, Amit Dvir and Chen Hajaj, 100,000 \$
- 6) 2019 – 2021, Israel Innovation Authority, Magnethon, Yifat ltd, Amit Dvir, Chen Hajaj, Uzi ben-Shalom, 100,000\$
- 7) 2016 – 2018, Israel Innovation Authority, Magnet, Nepton, Amit Dvir, SDN-WCP, 30,000 \$
- 8) 2015 – 2018 - Israel Innovation Authority, Magnet, InfoMedia, Amit Dvir, Classification and Enrichment of Encrypted Traffic, 80000\$
- 9) 2009 – 2014, Israel Innovation Authority, Magnet, NETHD, Boaz Ben Moshe, Amit Dvir, Algorithms for cooperative Multicast between ISPs, 250,000\$
- 10) 2011 – 2012, France Telecom, Michael Segal (BGU), Amit Dvir, Improved optical network placement, 1 year – 150000\$

## **Lectures and Presentations at Meetings and Invited Seminars not Followed by Published Proceedings**

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### **Seminar presentations at universities and institutions**

#### **Since arrived to Ariel University**

2025, Berno University.

2025, The KIOS Research and Innovation Center of Excellence, Cyprus

2020, Georgi Rakovski Military Academy, Bulgaria

2019, CrySyS Lab - BME-HIT Budapest

#### **Before arrived to Ariel University**

2012, Electrical and Electronics Engineering - Ariel University

2012, Software Engineering - the Jerusalem College of Engineering

2012, Computer Science - Interdisciplinary Center Herzliya

2009, CrySyS Lab - BME-HIT Budapest

## **Synopsis of research, including reference to publications and grants in above lists**

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My research lies at the intersection of systems and theory. I am motivated by problems presented by real life. My goal is to explore the properties of those systems and develop real-life algorithms. Communication network systems, such as ad hoc and Internet networks, intrigue me. I believe in a multidisciplinary research approach; I have been using tools and results from mathematics, machine learning, and computer science. I draw inspiration for problems from practice, apply a principled approach to develop solutions and transfer these solutions back to practice. I discover interesting and challenging practical problems through frequent discussions and meetings with industry professionals; in particular, I seek real-world network problems that present opportunities for breakthroughs by applying a principled approach. I then tackle these problems by applying principles, developing new methods, and transferring these solutions back to practice as working systems.